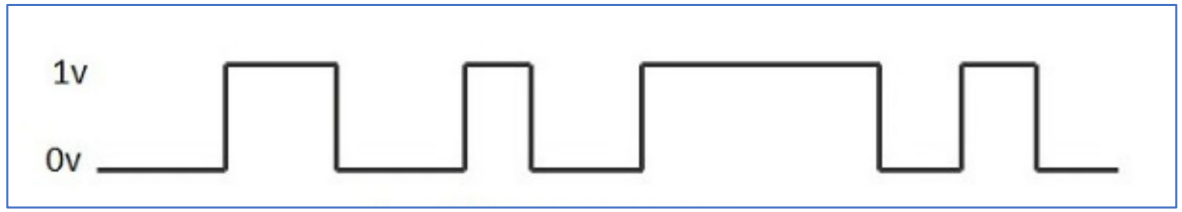
28 April 2026

University of Scranton Communications Systems

# Costas arrays and other high-correlation signals

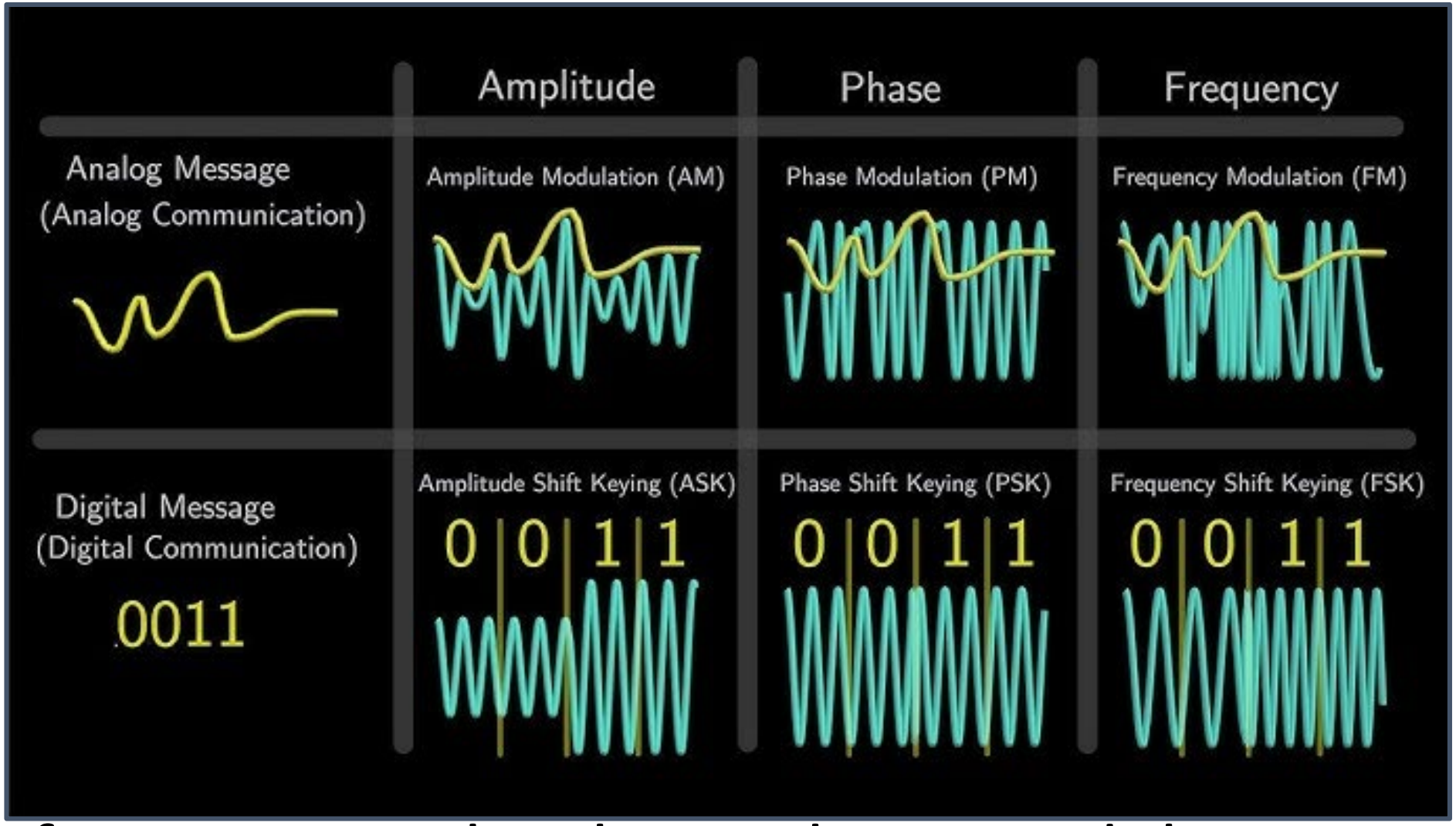
David Kazdan
Case Western Reserve University
Cleveland, Ohio USA



# Topics: Radar, spread-spectrum, signal patterns. Pulses, noise, Barker codes, Costas arrays.

# Correlation and ambiguity in noisy, pseudonoisy, and disorganized signals

- The communications channel may itself be the message
- Pulses and their echoes are adequate for simple radar and sonar
- Improvement: spread the pulse's energy in time
  - Barker codes
- Further improvement: spread the pulse's energy in both time and frequency
  - Costas arrays




Most of your course has been about modulation types. Encode information, send the signal, decode the information. That's communications!

# What's in common to all of these methods?

## Communications involves surprise!

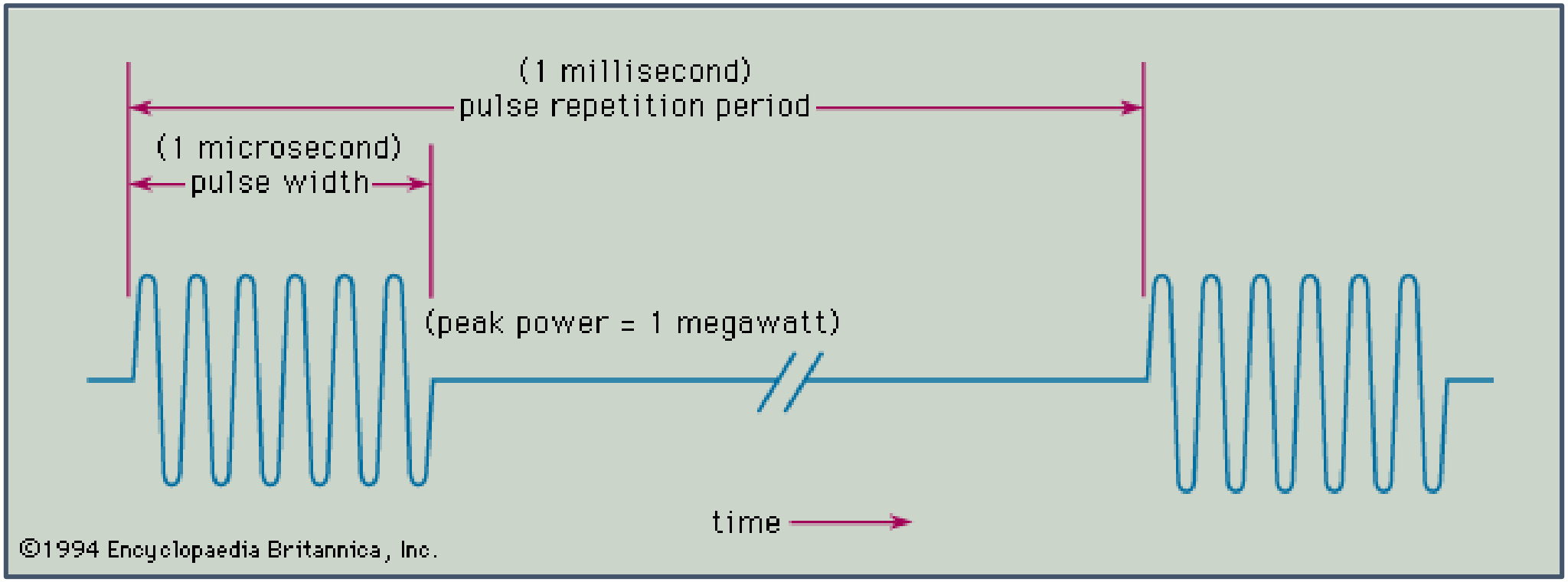
The frequency and bandwidth and modulation parameters are agreed on. The information content isn't known to the receiver; decoding gives a statistical view of the most likely message given the received signal the channel noise.

Sending an unknown signal may not be the goal.

In radar, the information is in the communications channel.

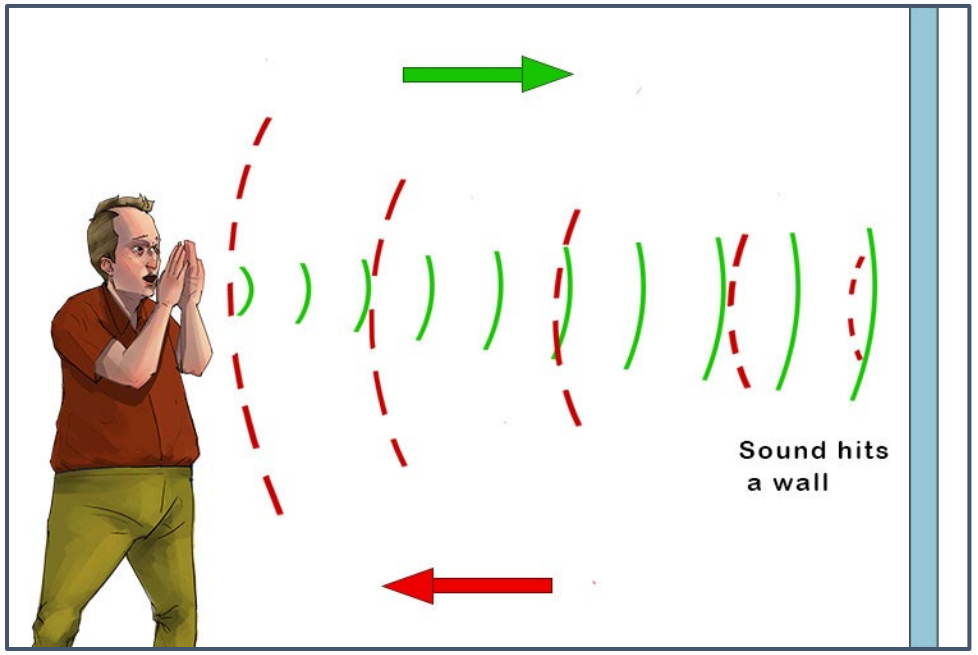
The signal *per se* is known ahead of time.

# Pulse radar sends out Morse “dits” and waits for their echo.



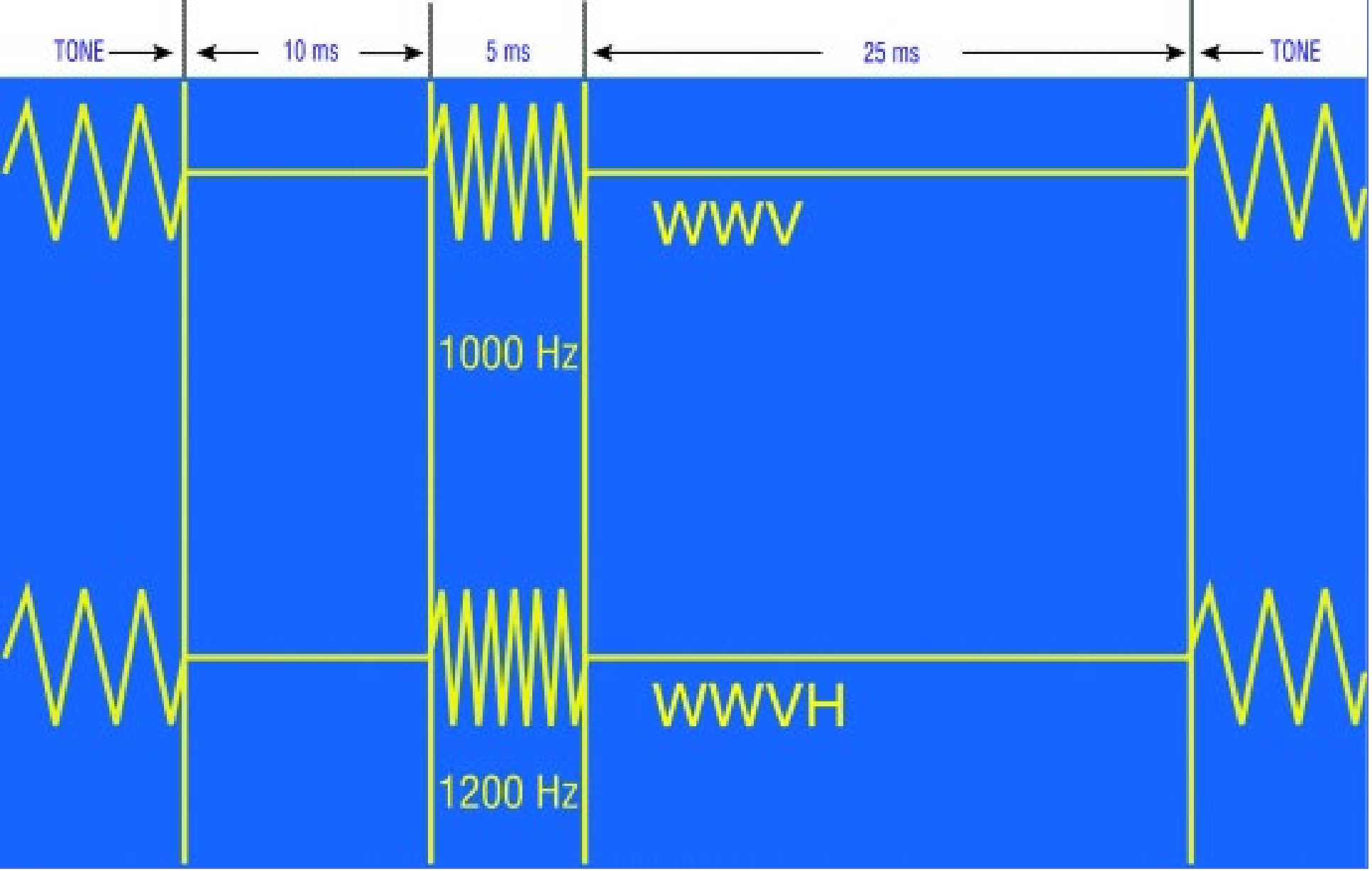

Megawatt!  You need that: The reflected signal is going to be weak by 1/r^4 (Friis equation)

# By the way, the echo is from an impedance mismatch. This is true in acoustic echos, also.




What's the acoustic boundary condition at the (rigid) wall?

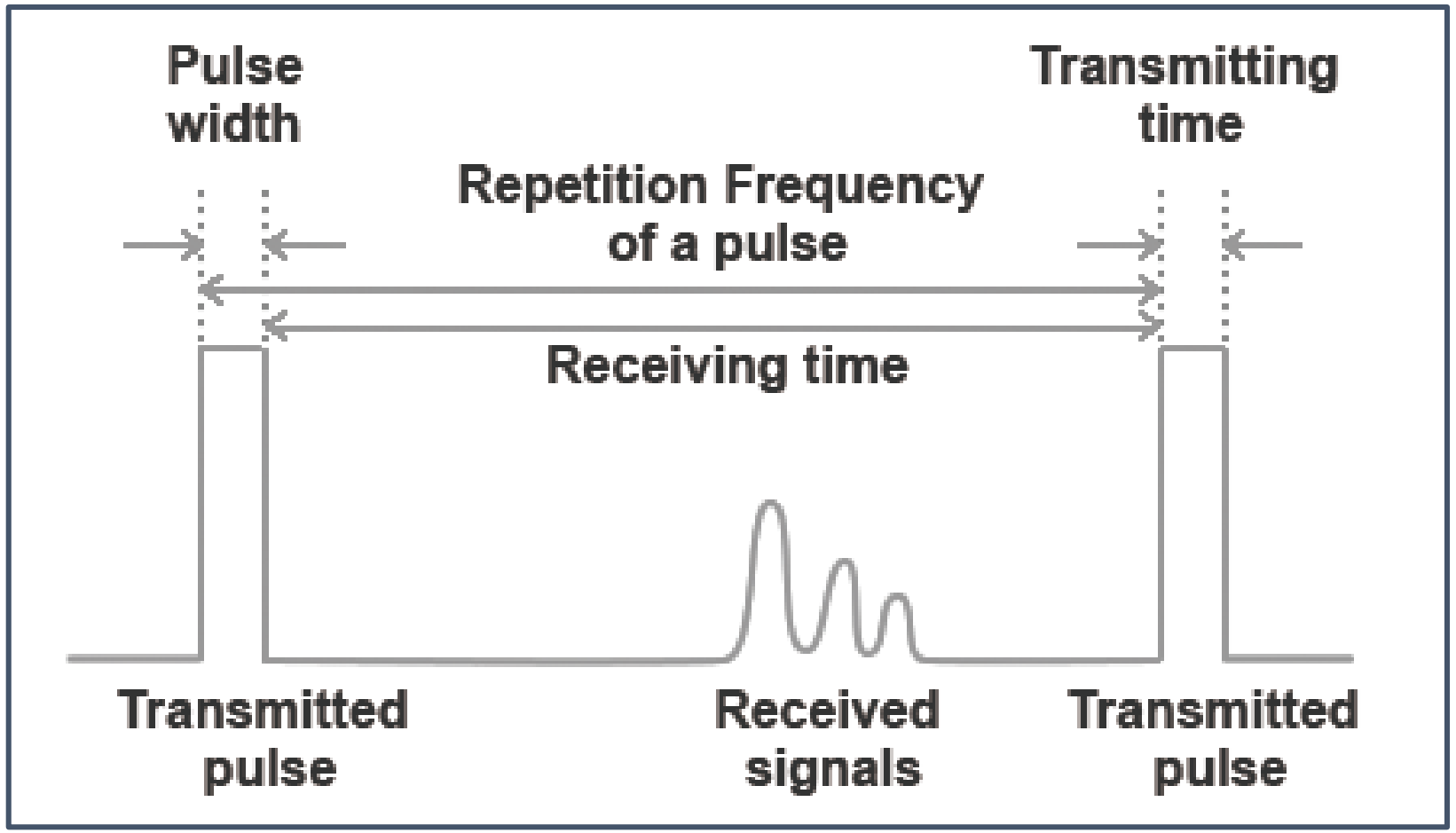
What is the RF boundary condition on the (conductive) aluminum skin of an airplane?




One-way radar? This is the timing signal from NIST's transmitter in Colorado.

WWV and WWVH second ticks are tightly defined and timed by caesium clocks.

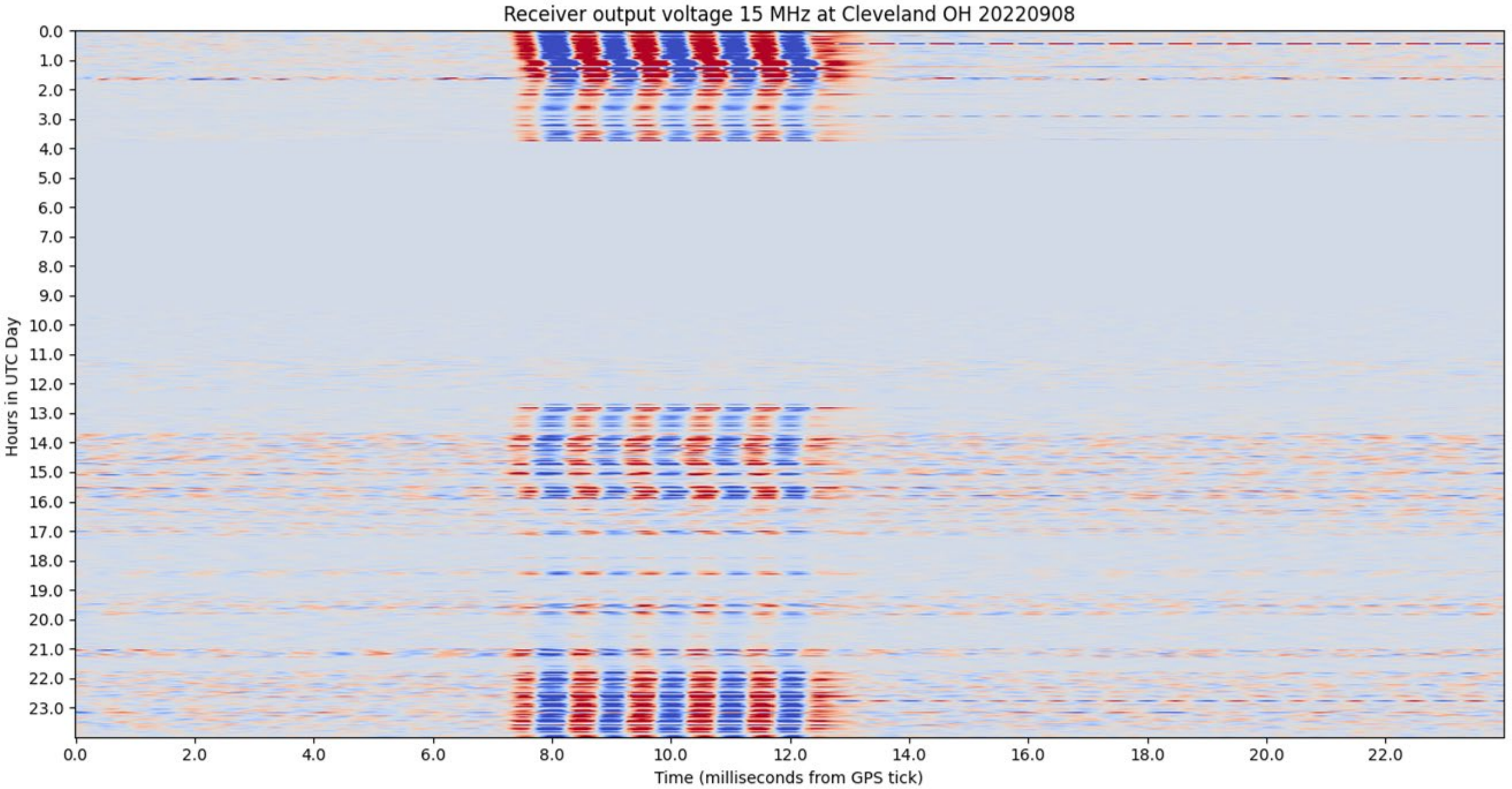
# The returned echo is likely not going to be “clean”




Noise might be from environment or from receiver amplifiers. That signal is weak.

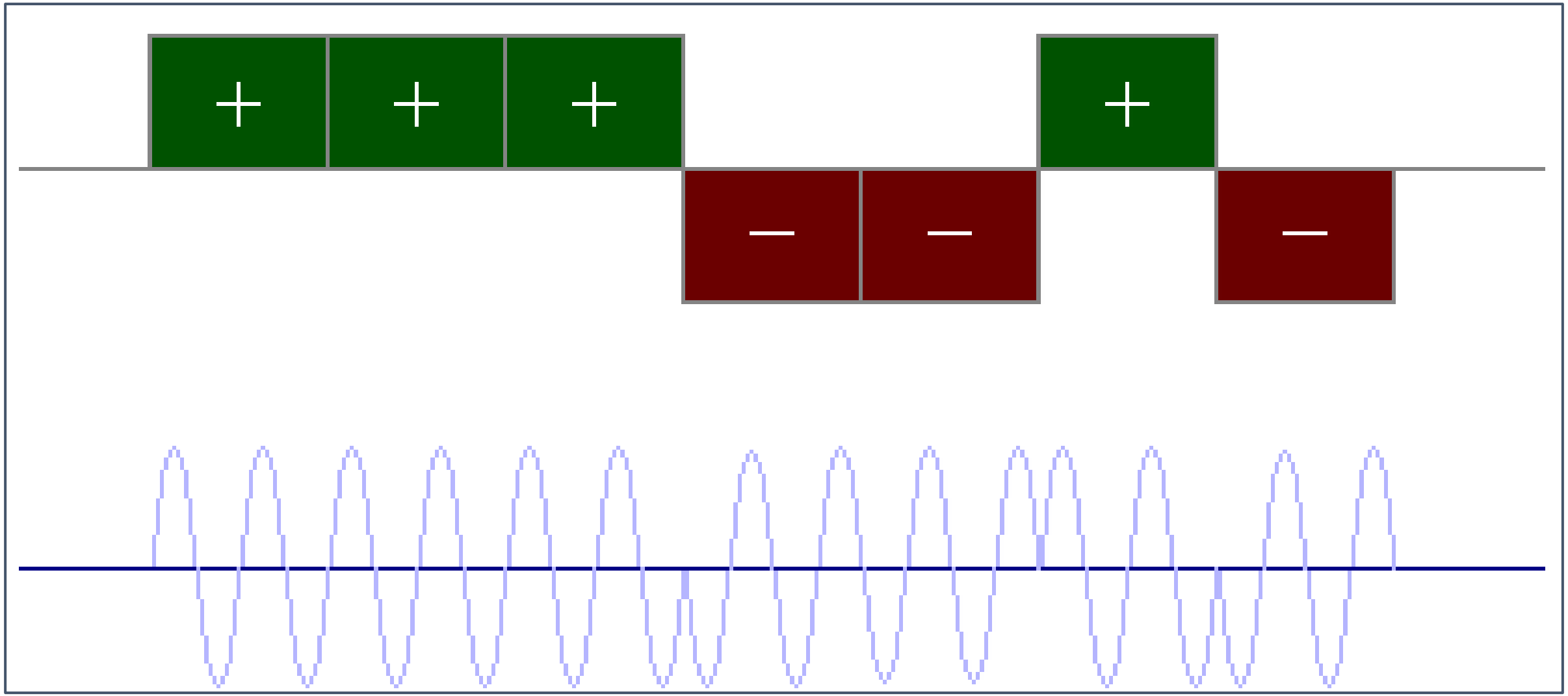
What might improve the statistics of the signal processing?

For on-off keying and phase-shift keying, Barker codes help.

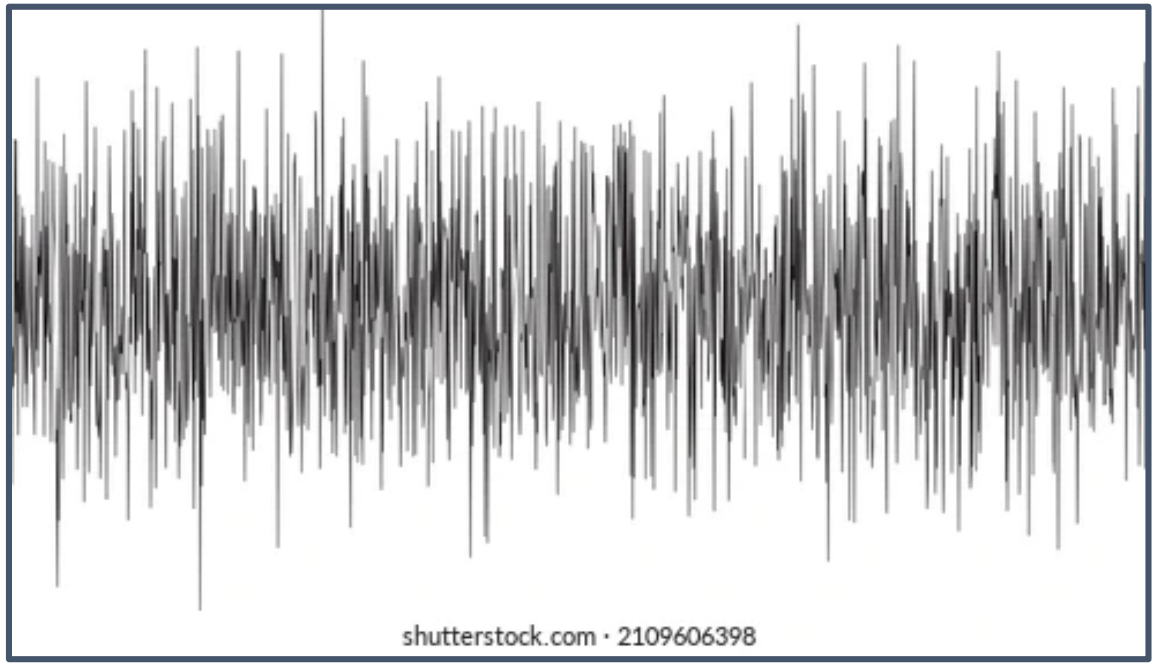


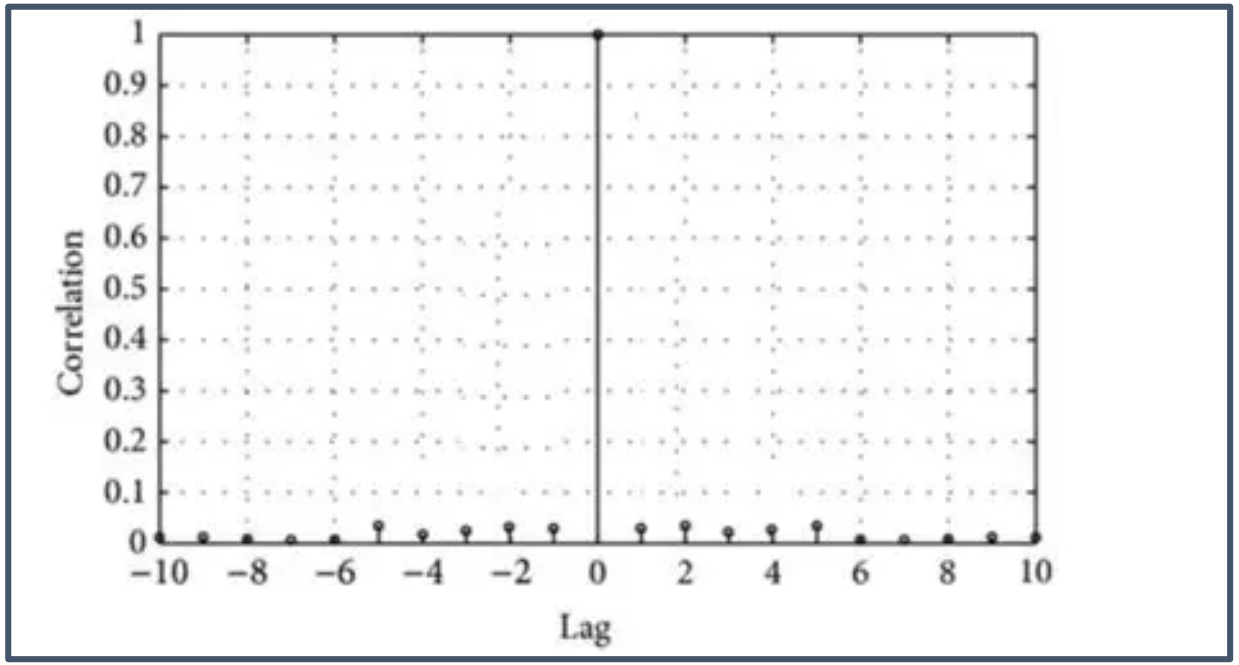

WWV second ticks as received: The sinusoid of the “tick” showing variable amplitude and delays over the UTC day. Delay varies with conditions.

# A pattern of pulses may help



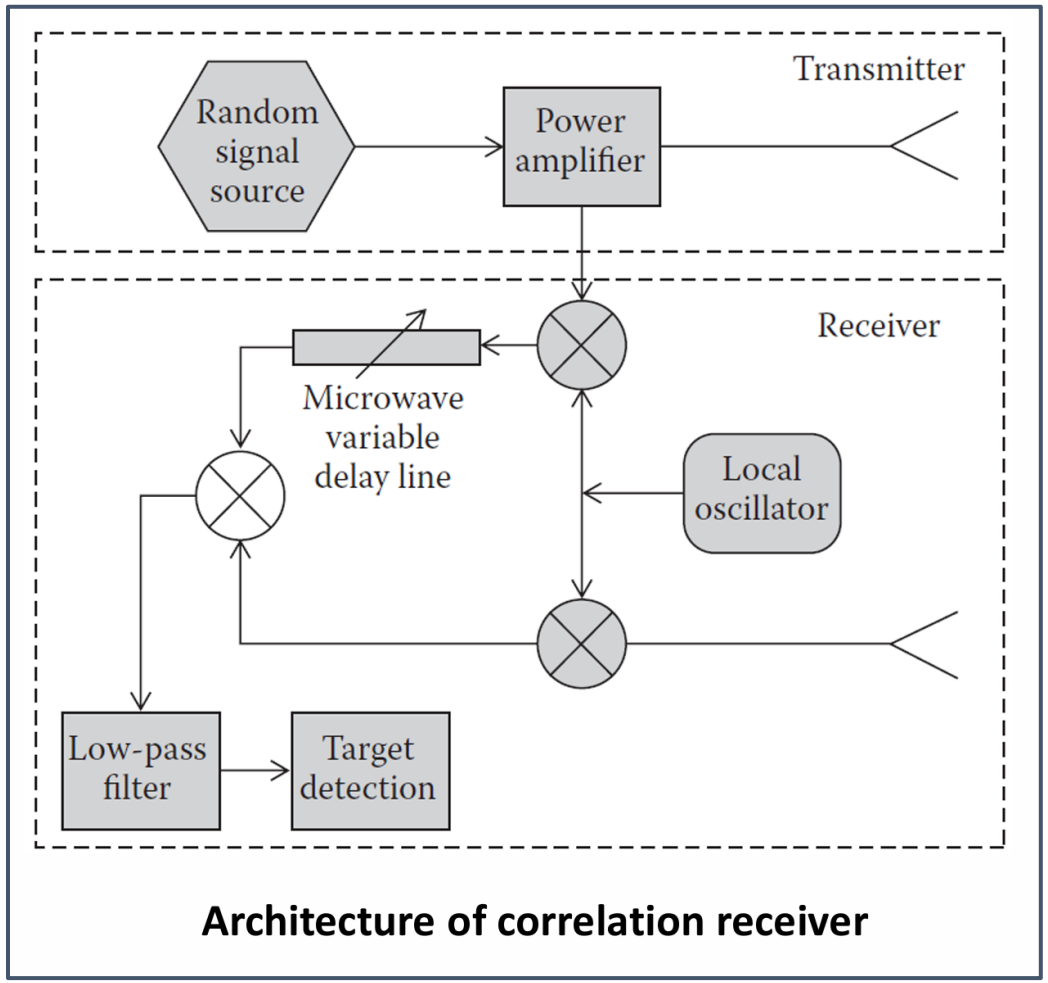
White noise has perfect autocorrelation. Problem: It also has infinite energy and infinite bandwidth. It's a mathematical starting point.



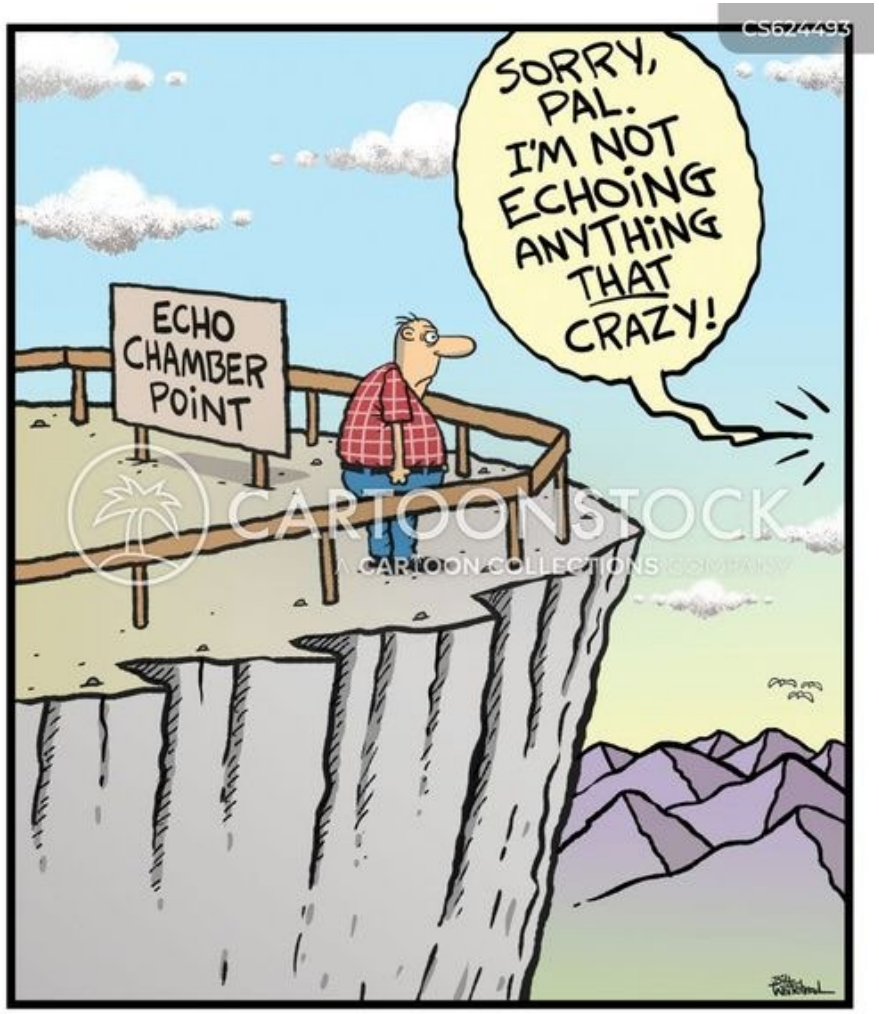




The correlation is zero everywhere except at perfect alignment, where it's infinite.

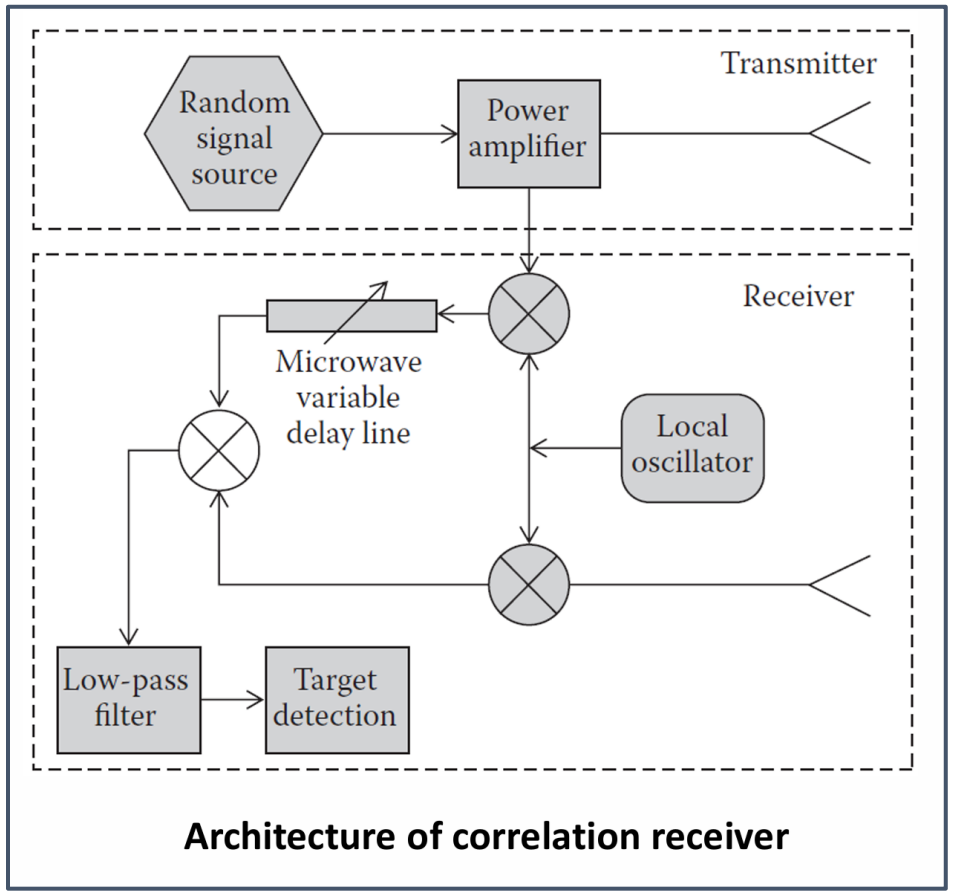
# We need a better solution than noise radar. Although: interesting slide set on that: Introduction to Random Signal Radar




**Architecture of correlation receiver**

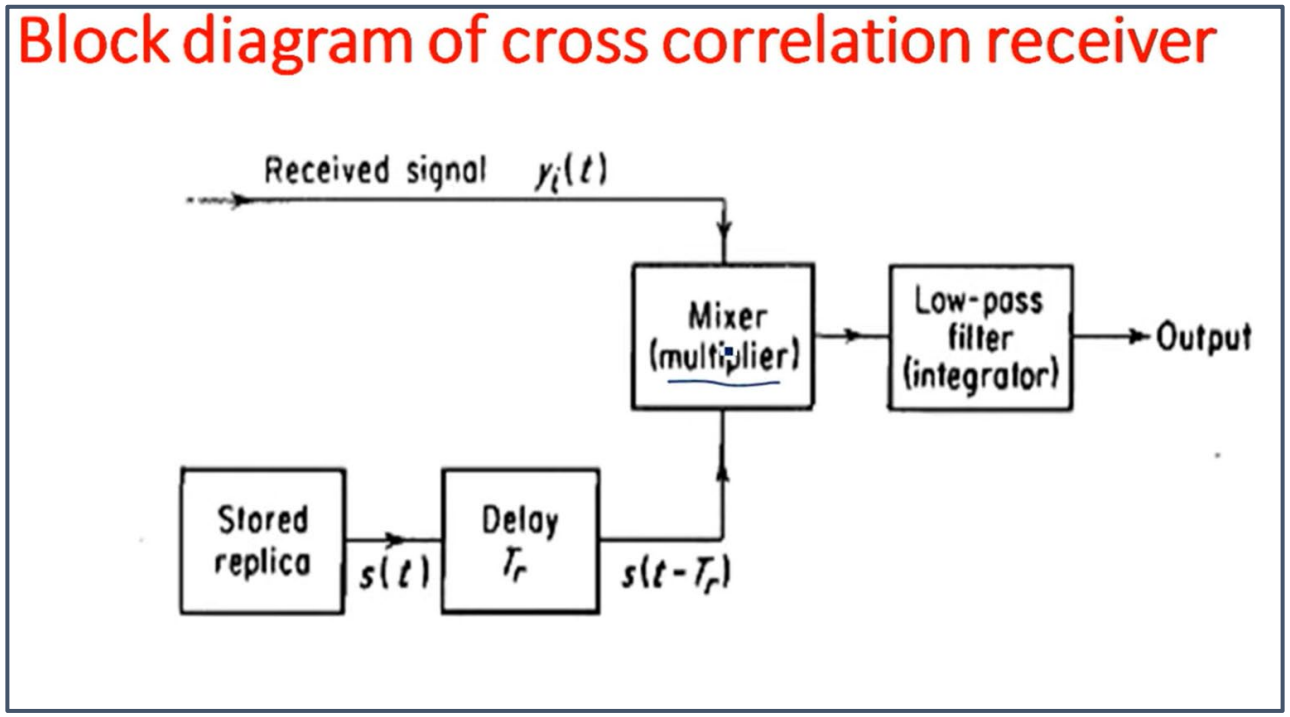


# What happens if the signal is not random, but pseudorandom, chaotic or disorganized?



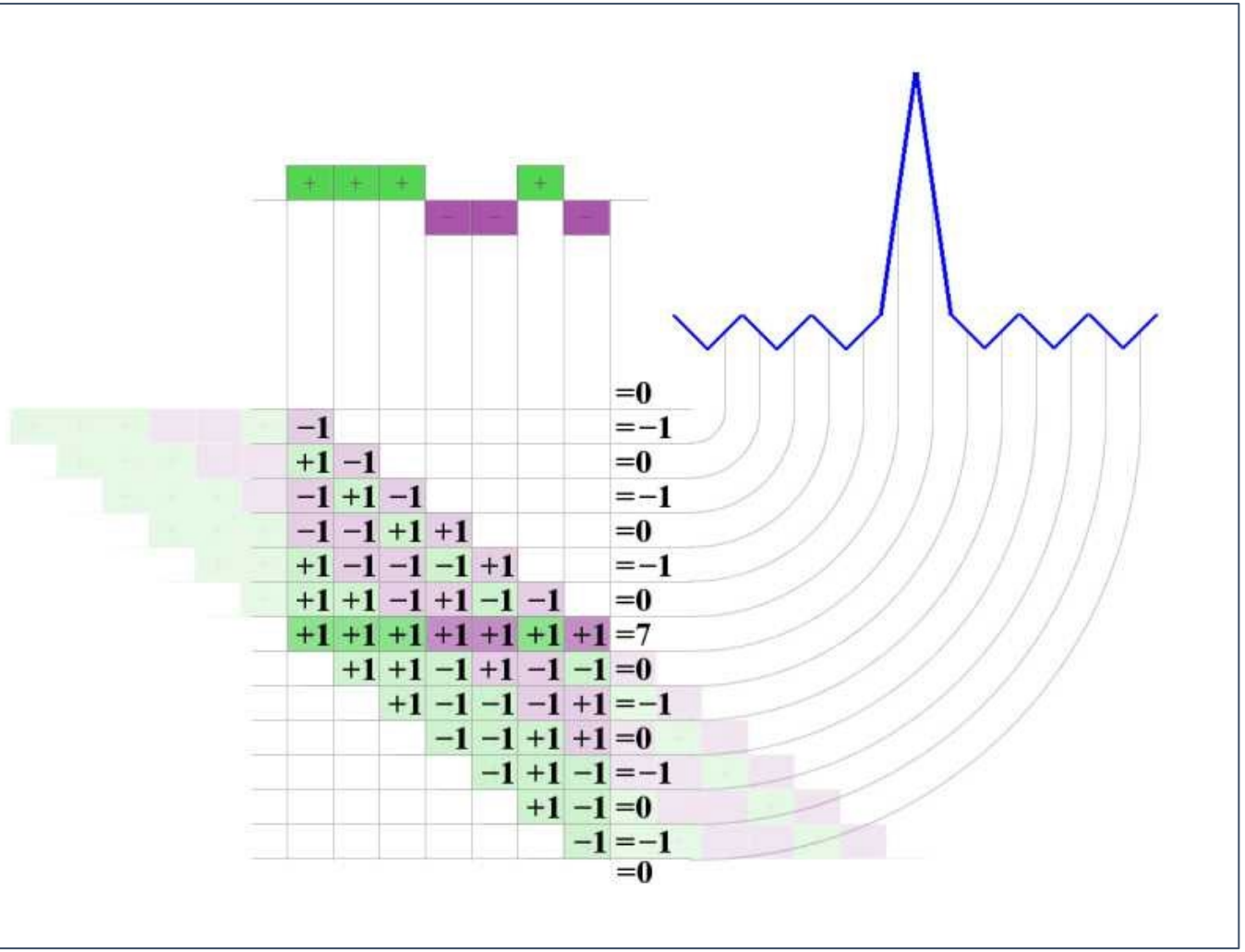

**Architecture of correlation receiver**

(repeated from last slide)
Need this for random signal.




This works for signal that is known: Pseudorandom, chaotic, disorganized.

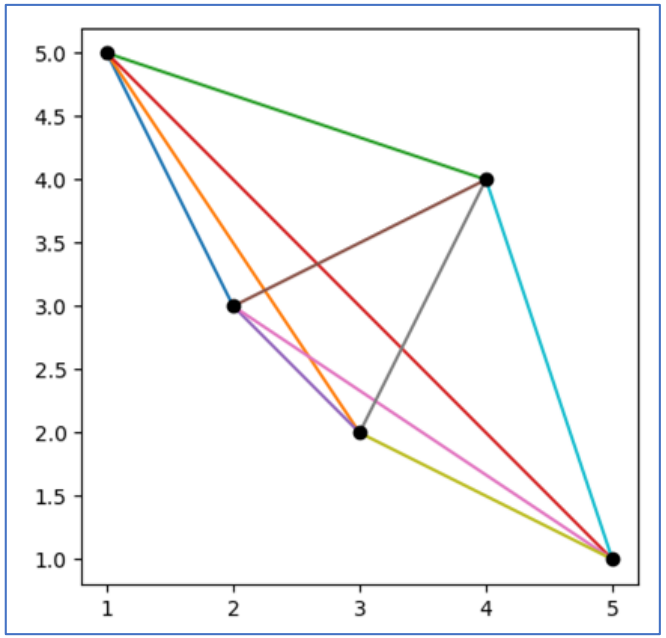
# An improvement for pulse radar: Barker codes. Here, pulses may be phase-inverted

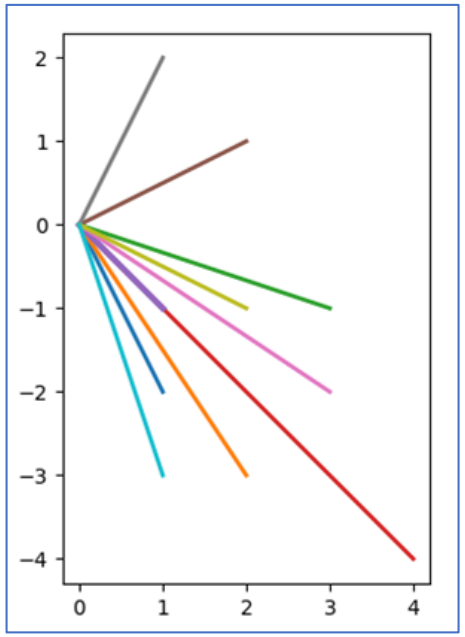



Simplest is just two pulses, evenly spaced. “click, click.”
Shows up in animal communications, too:

[Nature Might Pulse to a Universal Rhythm – 2 Beats Per Second : ScienceAlert]

It may be that you want Doppler radar, too.
Find range to target *and* radial velocity.
The rubric gets changed from “cross-correlation” to “ambiguity function,” but it’s the same thing.
Expand the 1-dimensional Barker code to 2-dimensional Costas arrays.

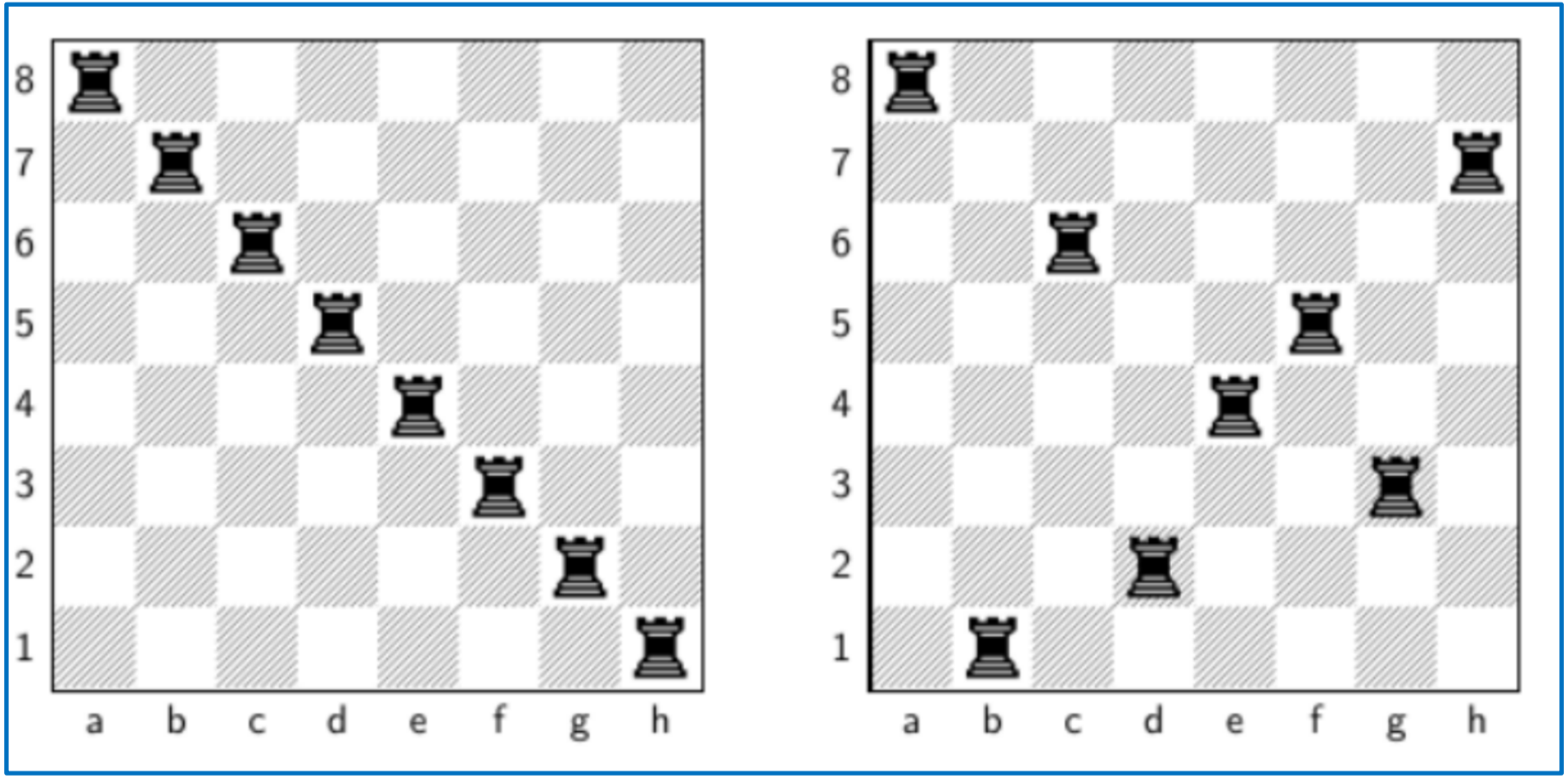


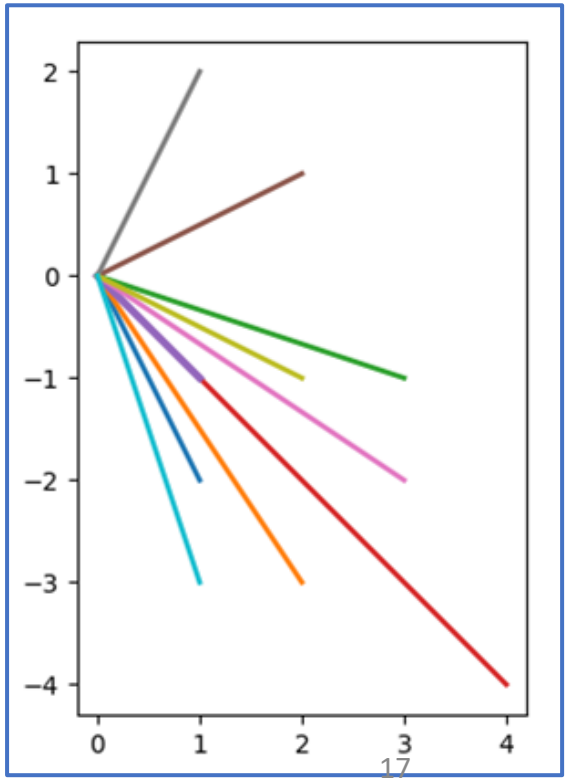



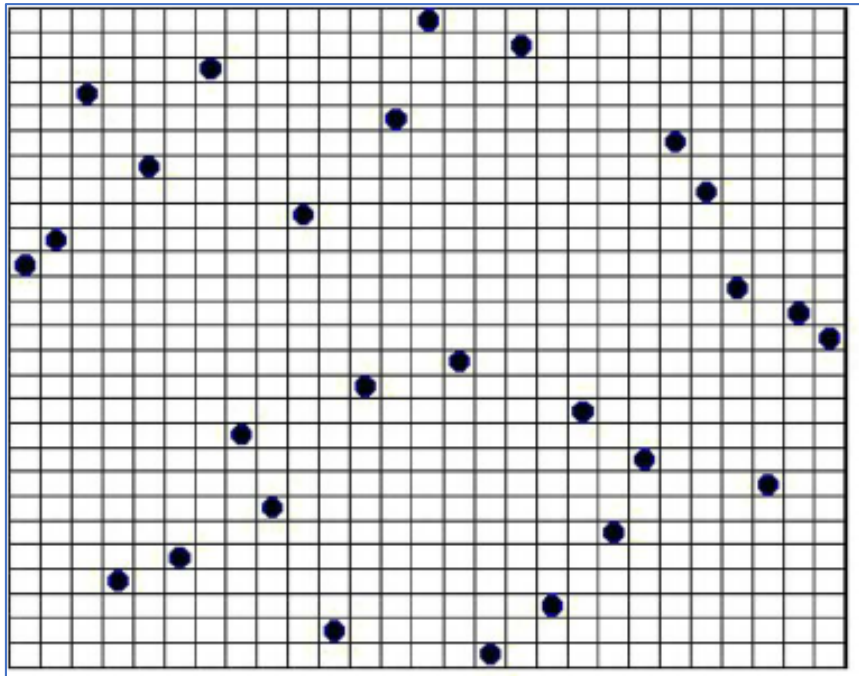

Vectors between some number of points in a square array, all different. Turns out to be similar to an *n*-rooks placement puzzle:

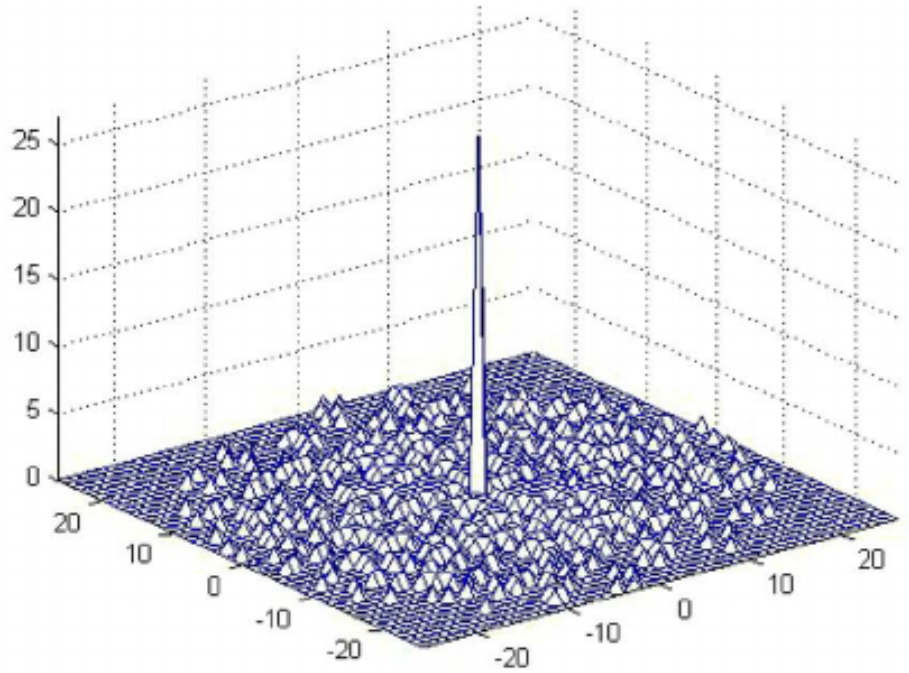
# Costas arrays are *maximally disorganized* but they are not random.

These are optimized schemes for radar and other pattern transmissions.



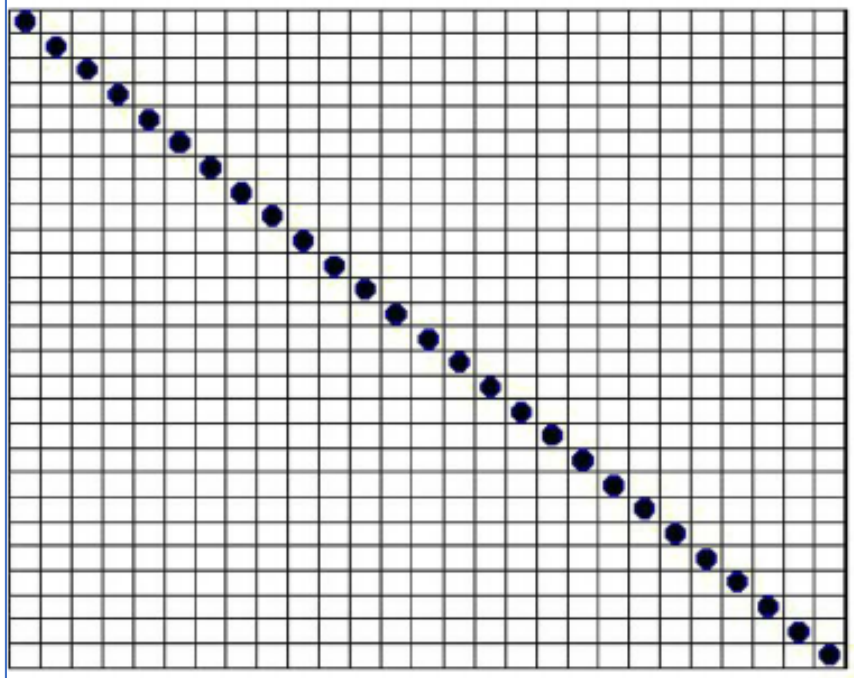



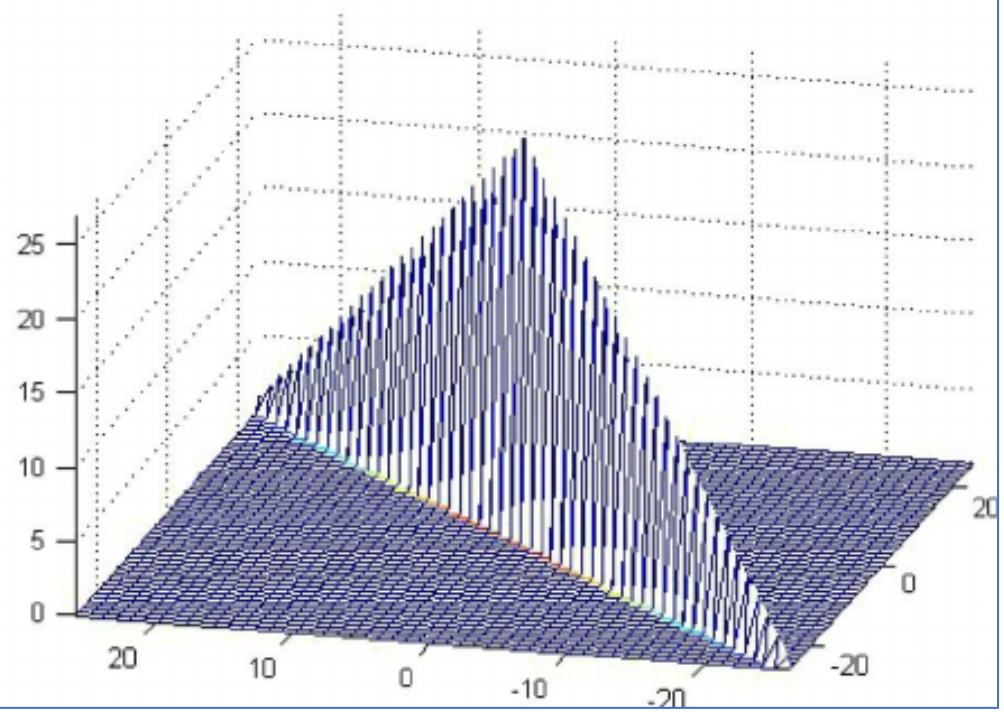





27-square Costas array, one frequency per time slot.

Autocorrelation is a “thumbtack.”






27-square discrete frequency sweep.

Autocorrelation is a “shark fin:” It has a peak, but much less definite than the Costas thumbtack.

# The two-dimensional optimized signal pattern is the Costas array. It is the synchronizing method for FT8 and its ilk. Recommended discussion here.

## FT8 SPECIFICATIONS

**Signal-to-Noise in 2500 Hz bandwidth:** –20 dB

**Modulation:** 8-tone Frequency Shift Keying

**Transmit duration:** 12.64 seconds

**Baud rate:** 6.25 bps

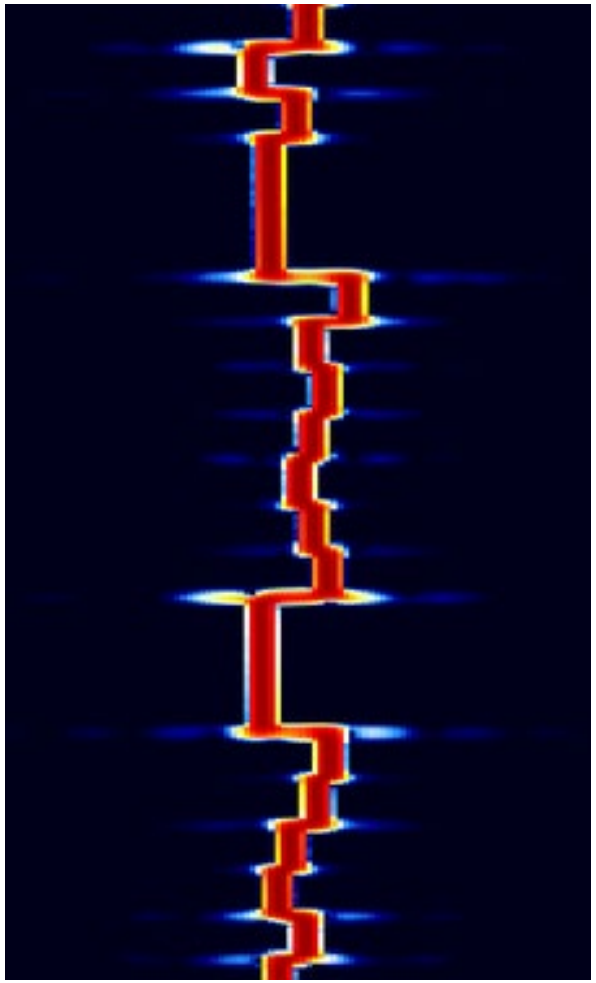
**Modulation bandwidth:** 50 Hz

**Forward Error Correction:** Low density parity check

**Synchronization:** 7x7 Costas Array using 26.5% of TX energy

FT8 depends on both time and frequency synchronization. It works with remarkably low-resolution equipment and unknown distances because of this method.

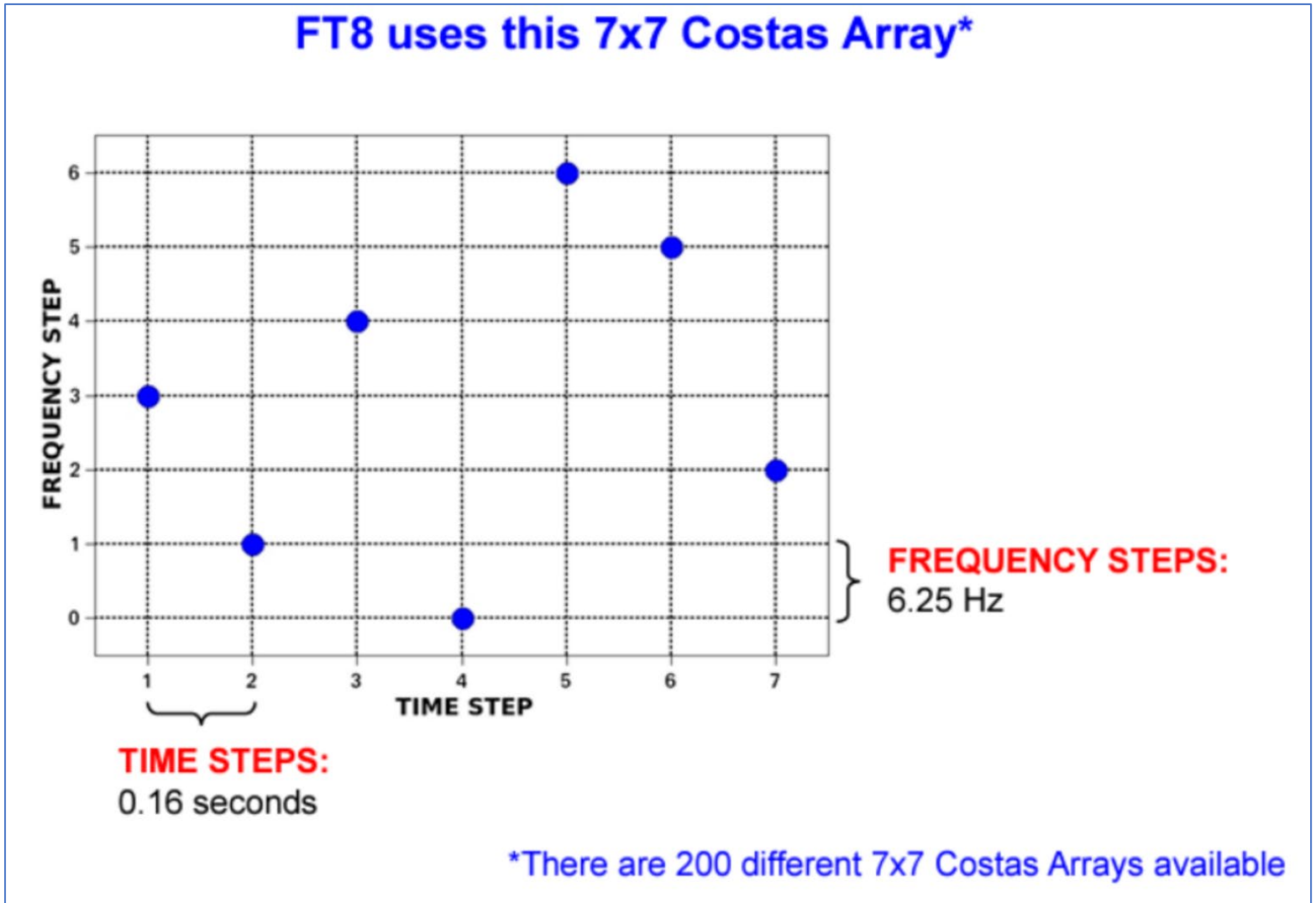
# FT8 uses 8-*m* o-FSK, so 3 bits per flag.



The signal has known time per flag.  It is sampled during that time, then the resulting data vector is put through Fourier transform.  There are 8 possible frequency bins; the one with maximum energy is has the maximum likelihood of being the correct decode.

The bins are labeled 0 through 7, corresponding to message 000 to 111 for that time slot.

# Synchronization in time and frequency: The receiving station knows what to expect.




FT8 data payload is only ~ ¾ of total transmission.

The rest is the synchronization .

That's the cost of doing business in bad S/N settings!

# Watch this space!

Several CWRU senior project groups are working on GNSS-synchronized Costas hardware for sounding the August 2026 eclipse. The work will extend into more general ionospheric characterization.

Undergraduate math majors are working on orthogonality of same-order arrays, to investigate their use as a flagging code.

After the eclipse, we anticipate amateur radio experiments in ionospheric sounding using this method. With successes there, we can propose a WWV/H implementation.

TNX ES 73 DE AD8Y

send an email to dxk10@cwru.edu to be included in the library.